Experiments

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Molecular Cloning

This chapter covers the construction workflow of all recombinant plasmids, including:

1. Backbone linearization by inverse PCR

2. Target gene fragment amplification (with homology arms)

3. Agarose gel electrophoresis and gel extraction

4. In-homologous recombination ligation

5. E. coli DH5α competent cell transformation

6. Colony PCR verification and plasmid extraction

1.1 Backbone Linearization by Inverse PCR

The pCAMBIA1302 vector was used as the backbone, linearized by inverse PCR. Two backbone variants were designed:

1. 35S group: retaining the CaMV 35S promoter, primers p1302-IF / p1302-IR

2. *RBCS2* group: replacing 35S with the *RBCS2* promoter, primers p1302-IF / p1302-35s-IR

PCR reaction (50 µL):

Reagent	Volume
KOD One™ PCR Master Mix (2×)	25 µL
Forward Primer (10 µM)	1.5 µL
Reverse Primer (10 µM)	1.5 µL
pCAMBIA1302 Template	1 µL

Table 1. Inverse PCR reaction system

PCR program:

Step	Temperature	Time	Cycles
Initial Denaturation	98℃	3 min	1×
Denaturation	98℃	10 s	
Annealing	58–66℃ (Touchdown)	5 s	
Extension	68℃	2 min	34×
Final Extension	68℃	5 min	1×

Table 2. Inverse PCR program

1.2 Target Gene Amplification (with Homology Arms)

Target gene fragments were amplified from glycerol stock plasmid templates, with 15–20 bp homology arms to the backbone ends introduced at both primer ends. The following fragments were amplified:

1. *CCD1* series: C1-*CCD1*, C3-*CCD1* (no ATG + signal peptide), D1-*CCD1*

2. *CitCCD4* series: C2-*CitCCD4*, D2-*CitCCD4*

3. *RBCS2* + SP (signal peptide) fragment: used for C1/C2/C3 constructs

PCR program:

Table 3. Target gene amplification program

Step	Temperature	Time	Cycles
Initial Denaturation	98℃	3 min	1×
Denaturation	98℃	10 s	34×
Annealing	55–60℃	5 s	
Extension	68℃	20 s	
Final Extension	68℃	3 min	1×

1.3 Agarose Gel Electrophoresis and Gel Extraction

1. Prepare 0.8% (large fragment backbone) or 1.0% (small target fragment) agarose gel with 4S GelRed nucleic acid stain

2. Run electrophoresis at 120 V in 1× TAE buffer for 25–30 min

3. Excise target bands under UV light; purify using Omega / Tiangen gel extraction kit

4. Measure recovered concentration with NanoDrop

1.4 In-Homologous Recombination with CloneExpress

Seamless cloning was performed using the Vazyme ClonExpress Ultra One Step Cloning Kit. Reaction system (10 μL):

1. 2× CloneExpress Mix: 5 μL

2. Linearized backbone: insert added at 1:2–1:5 molar ratio

3. ddH_2O to 10 μL

4. Incubate at 50°C for 30 min

Table 4. Five recombinant plasmids constructed:

ID	Promoter	Target Gene
C1	*RBCS2*	*CCD1*
C2	*RBCS2*	*CitCCD4*
C3	*RBCS2*	*CCD1* (with signal peptide)
D1	35S	*CCD1*
D2	35S	*CitCCD4*

1.5 E. coli DH5α/Stbl3 Transformation

1. Thaw 100 μL DH5α or Stbl3 competent cells on ice

2. Add all homologous recombination product, mix gently by flicking, incubate on ice for 30 min

3. Heat shock at 42°C water bath for 90 s, then immediately chill on ice for 2 min

4. Add 700 μL antibiotic-free LB liquid medium, recover at 37°C, 200 rpm for 60 min

5. Centrifuge at 4000 rpm for 3 min, discard part of supernatant, resuspend pellet in 100 μL

6. Plate onto LB solid plates containing kanamycin (50 μg/mL), incubate inverted at 37°C overnight

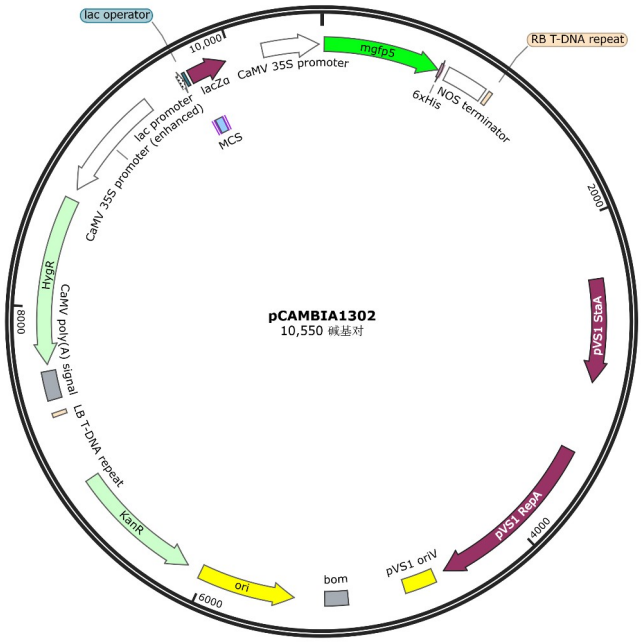


Figure 1. Backbone plasmid pCAMBIA1302 map (11,849 bp).

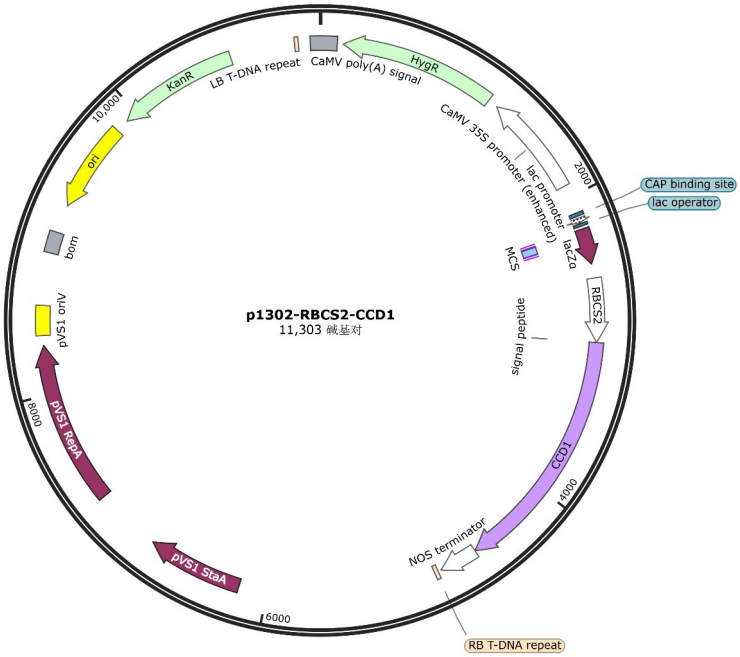


Figure 2. p1302-*RBCS2-CCD1* recombinant plasmid map (11,303 bp).

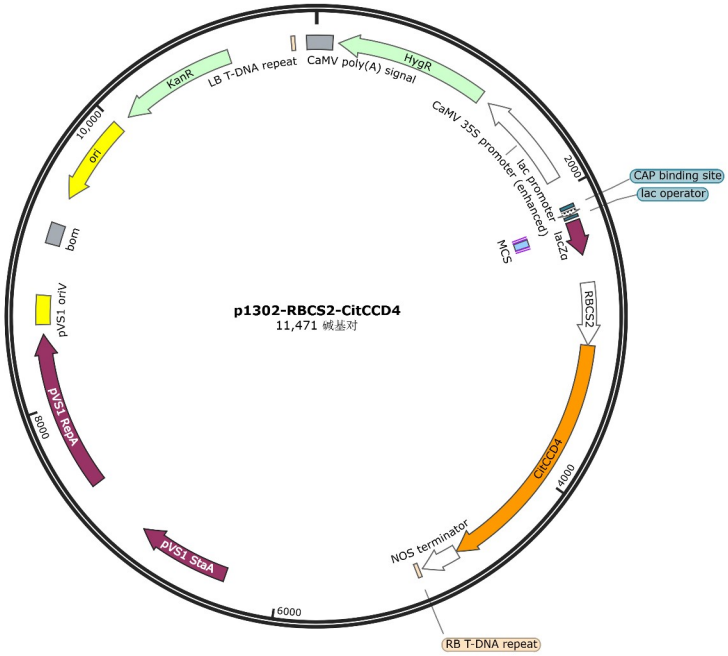


Figure 3. p1302-*RBCS2*-*CitCCD4* recombinant plasmid map.

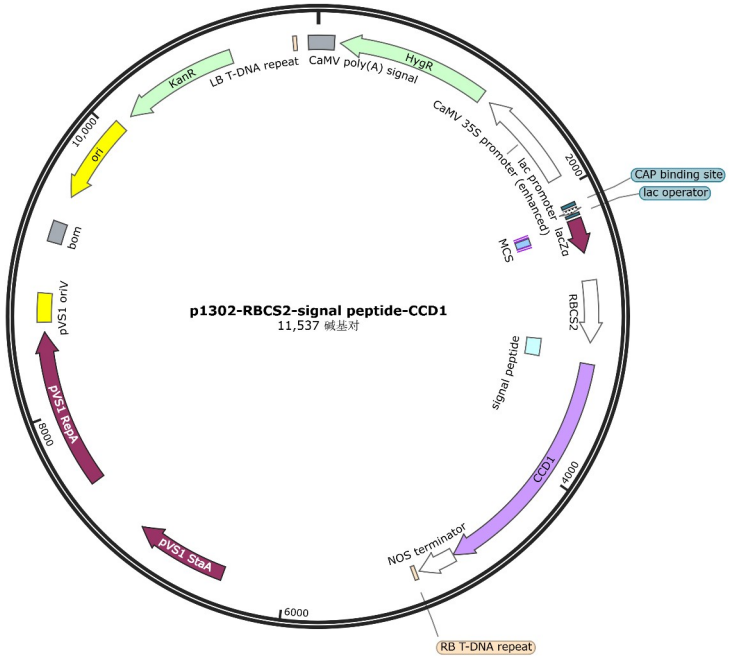


Figure 4. p1302-*RBCS2*-SP-*CCD1* recombinant plasmid map (with chloroplast transit peptide).

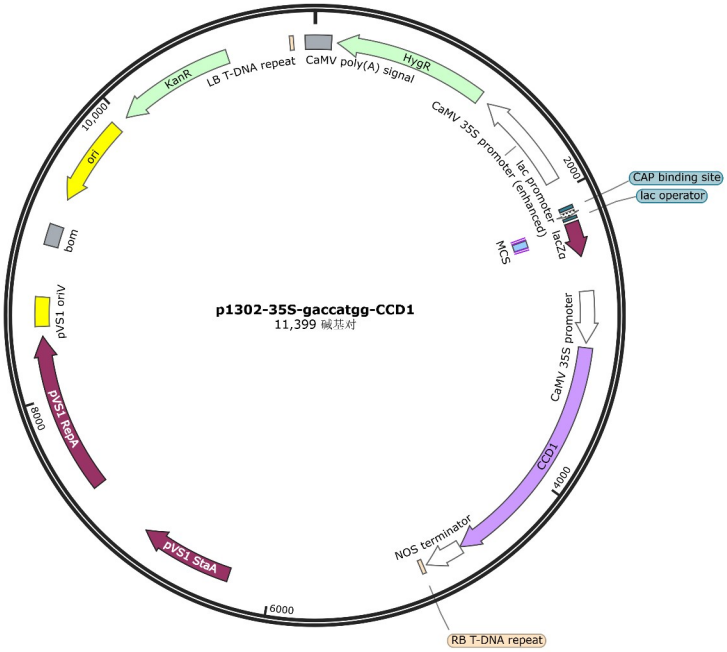


Figure 5. p1302-35S-*CCD1* recombinant plasmid map (driven by 35S promoter).

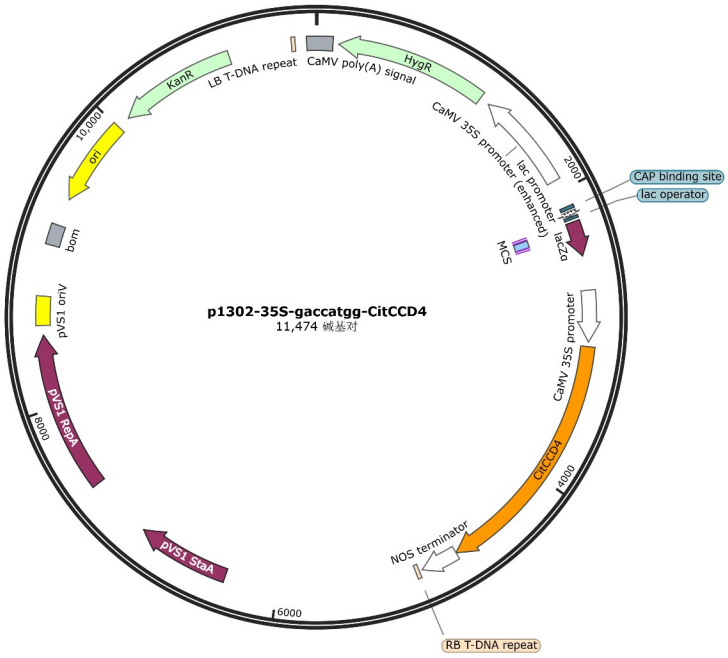


Figure 6. p1302-35S-*CitCCD4* recombinant plasmid map (driven by 35S promoter).

1.6 Colony PCR Screening and Plasmid Extraction

1. Pick single colonies, resuspend in 10 µL sterile water as PCR template

2. Perform colony PCR with gene-specific primers (system as in 1.2)

3. Inoculate positive clones into 5 mL LB liquid medium with kanamycin, grow at 37°C, 200 rpm overnight

4. Extract plasmids with Tiangen miniprep kit, measure concentration with NanoDrop

5. Send positive plasmids to a sequencing company for Sanger sequencing verification

Algal Transformation

This chapter covers the culture and transformation workflow of *Dunaliella salina*, including:

1. DM medium preparation and algal expansion culture

2. Transformation by osmotic shock method

3. Post-transformation recovery culture

2.1 Medium Preparation and Algal Culture

Modified DM medium (1.5 M NaCl, 1 L) formulation:

Component	Amount
NaCl	87.69 g
$NaNO_3$	0.42 g
$NaH_2PO_4 \cdot 2H_2O$	15.60 mg
$MgSO_4 \cdot 7H_2O$	1.23 g
KCl	74.00 mg
$CaCl_2 \cdot 2H_2O$	44.00 mg
10× A5 solution	100 µL
$NaHCO_3$	0.84 g
Fe·EDTA solution	1 mL

Table 5. Modified DM medium formulation (1.5 M NaCl, 1 L)

Culture conditions: 26°C, 50 µmol photons·m^{-2}·s^{-1} light intensity, 16 h light / 8 h dark photoperiod, static culture.

2.2 Transformation by Osmotic Shock

Host strain:

Dunaliella salina FACHB-435, obtained from the Freshwater Algae Culture Collection at the Institute of Hydrobiology (FACHB), CAS.

Transformation steps:

1. **Take log-phase algae (OD_{680} ≈ 0.5–1.0, density ~10^5–10^6 cells/mL), centrifuge at 5000 rpm for 5 min to collect pellet**
2. **Wash 3 times with fresh DM medium to thoroughly remove residual metabolites**
3. **Adjust cell concentration to 10^6 cells/mL, pre-adapt in 1 M NaCl medium for 8 h**
4. **Take 1 mL pretreated cell suspension, centrifuge and discard supernatant**
5. **Immediately resuspend pellet in 1 mL 0.1 M NaCl hypotonic medium**
6. **Add 5.46 μg plasmid DNA and 15 μL 0.1% Triton X-100**
7. **Vortex vigorously for 60–120 s (recommended: 90 s)**
8. **Immediately transfer to 30 mL fresh DM medium, allow recovery at 26°C static for 48 h**

Three batches of transformations were completed: Aug 17 (batch 1, 10 plasmid combinations), Aug 21 (batch 2), Aug 30 (batch 3, one each of C1/C2/C3/D1/D2).

Screening & Verification

This chapter covers the screening and molecular identification workflow of transformants:

1. **Hygromycin concentration gradient pre-experiment**
2. **Genomic DNA extraction from *D. salina***
3. **Genomic PCR verification of transformants**
4. **Sequencing verification**

3.1 Hygromycin Concentration Gradient Screening

Since pCAMBIA1302 carries the *hptII* (hygromycin phosphotransferase) resistance gene, hygromycin B was used as the selection antibiotic.

1. **Pre-experiment stage (Aug 22): wide gradient of 100, 200, 300, 600, 800 mg/L in 0.5 M NaCl DM medium**
2. **Refinement stage (Aug 28): based on the reported glucose modulation effect, set a gradient of 6.25, 12.5, 25, 50, 75, 100, 200 mg/L with 10 mM glucose co-treatment**
3. **Microscopic examination of cell viability at each concentration on Sep 5, recording live cell density gradient changes**

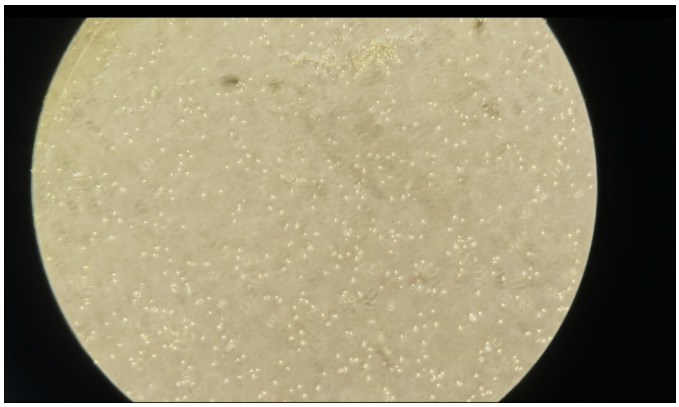


Figure 7. Microscopic view of hygromycin gradient with glucose co-treatment (12.5 mg/L).

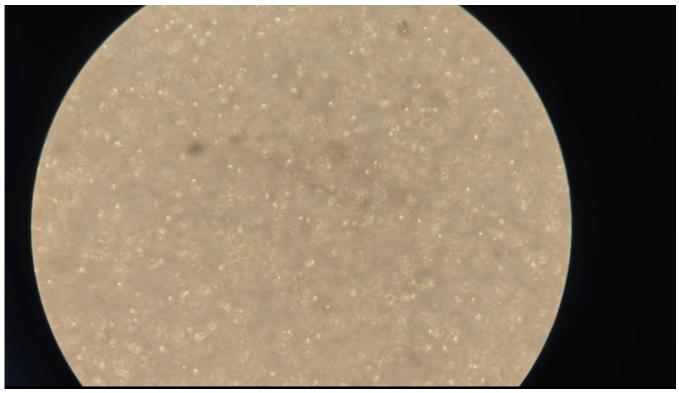


Figure 8. Microscopic view of hygromycin gradient (25 mg/L).

3.2 Genomic DNA Extraction from *D. salina*

Using Tiangen plant genomic DNA extraction kit (spin column method):

1. Take 10 mL cultured algal suspension, centrifuge to collect algal pellet

2. Grind to fine powder in liquid nitrogen, transfer to 2 mL centrifuge tube

3. Add 400 µL GP1 buffer (with RNase A), vortex to mix

4. Incubate in 65°C water bath for 10–30 min, invert to mix several times

5. Add 400 µL GP2 buffer, mix thoroughly

6. Centrifuge at 12000 rpm for 5 min, transfer supernatant to CB3 adsorption column

7. Wash twice with GD buffer, spin empty column for 2 min to remove ethanol

8. Elute with 50–100 µL TE buffer, incubate at room temp for 2 min, then centrifuge to collect DNA

3.3 Genomic PCR Verification of Transformants

Using extracted *D. salina* genomic DNA as template, perform PCR with gene-specific primers:

1. D1-*CCD1*-F / *CCD1*-R: verify D1 construct

2. *RBCS2*-F / *CCD1*-R: verify C1/C2/C3 constructs

3. D2-*CitCCD4*-F / *CitCCD4*-R: verify D2 construct

4. C2-*RBCS2*-F / *CitCCD4*-R: verify C2 construct

PCR system (25 μL): KOD One™ Master Mix 12.5 μL + forward and reverse primers 0.75 μL each + genomic template 1 μL + ddH_2O to 25 μL. Program as follows:

Step	Temperature	Time	Cycles
Initial Denaturation	98℃	3 min	1×
Denaturation	98℃	10 s	
Annealing	65℃	5 s	
Extension	68℃	20 s	34×
Final Extension	68℃	3 min	1×

Table 6. Genomic PCR verification of transformants

Metabolomics Analysis

To evaluate metabolic reprogramming in transformants, widely targeted metabolomics analysis was performed on three sample groups (C1, C3, and WT wild type).

Group	Sample	Host	Construct
C1 group	C1	*Dunaliella salina* FACHB-435	*RBCS2-CCD1*
C3 group	C3	*Dunaliella salina* FACHB-435	*RBCS2*-SP-*CCD1*
WT group	WT	*Dunaliella salina* FACHB-435	Wild Type Control

Table 7. Metabolomics sample information

This chapter covers sample preparation, metabolite extraction, UPLC-MS/MS widely targeted detection, carotenoid targeted quantification, and data preprocessing with statistical analysis, as detailed below:

4.1 Sample Preparation

1. Centrifuge to collect cultured algal cells, discard supernatant;

2. Lyophilize algal cells in a freeze dryer (Scientz-100F) under vacuum for 63 h;

3. Grind with a mill (MM 400, Retsch) at 30 Hz for 1.5 min (1 min for carotenoid targeted detection) to a fine powder; store at −80°C until use;

4. For analysis, weigh 30 mg algal powder for widely targeted metabolomics and 50 mg for carotenoid targeted detection.

4.2 Metabolite Extraction (Widely Targeted Metabolomics)

1. Add 1500 µL of pre-chilled (−20°C) 70% methanol aqueous extraction solution with internal standard to 30 mg algal powder, mix thoroughly;

2. Vortex for 30 s every 30 min, for a total of 6 cycles;

3. Centrifuge at 12,000 rpm for 3 min, collect supernatant, filter through a 0.22 µm microporous membrane;

4. Transfer filtrate to an autosampler vial for UPLC-MS/MS analysis.

Note: The internal standard solution was prepared by dissolving 1 mg standard in 1 mL 70% methanol water to make a 1000 µg/mL stock, then diluting with 70% methanol water to 250 µg/mL.

4.3 UPLC-MS/MS Detection (Widely Targeted Metabolomics)

Data acquisition was performed using an ultra-high performance liquid chromatography (ExionLC™ AD) coupled with tandem mass spectrometry (MS/MS) system.

(1) Chromatographic conditions

1. Column: Agilent SB-C18 (1.8 µm, 2.1 mm × 100 mm);

2. Mobile phase: A = ultrapure water (0.1% formic acid), B = acetonitrile (0.1% formic acid);

3. Elution gradient as shown in Table 8;

4. Flow rate 0.35 mL/min, column temperature 40°C, injection volume 2 µL.

Time (min)	Mobile Phase B Ratio
0.00	5%
9.00	95%
10.00	95%
11.10	5%
14.00	5%

Table 8. UPLC gradient elution program (widely targeted)

Note: 0–9 min linear gradient (5%→95%); 9–10 min hold at 95%; 10.00–11.10 min ramp back to 5%; 11.10–14.00 min equilibration at 5%.

(2) Mass spectrometry conditions

1. Ion source: electrospray ionization (ESI), source temperature 500°C;

2. Spray voltage: 5500 V positive mode, −4500 V negative mode;

3. Ion source gases GS1, GS2, and curtain gas (CUR) set to 50, 60, and 25 psi respectively; collision-activated dissociation (CAD) set to high;

4. Scan mode: triple quadrupole multiple reaction monitoring (MRM); collision gas (nitrogen) set to medium. MRM ion pairs were monitored according to eluting metabolites; declustering potential (DP) and collision energy (CE) were individually optimized.

4.4 Qualitative and Quantitative Principles

1. Qualitative: metabolites were identified based on the in-house MWDB database combined with MS/MS spectral information. Isotope signals, K^+/Na^+/NH_4^+ adducts, and large fragment ion duplicates were removed during analysis;

2. Quantitative: characteristic ion signals were acquired in triple quadrupole MRM mode; chromatographic peaks were integrated and calibrated, with peak area representing relative metabolite abundance.

4.5 Carotenoid Targeted Quantification

Targeted quantification of carotenoids and their esters (47 compounds total, including β-carotene, β-cryptoxanthin, zeaxanthin, β-citraurin, etc.) was performed for the target metabolic pathway.

（1）Sample Preparation

1. Add 0.5 mL of n-hexane/acetone/ethanol (1:1:1, v/v/v) containing 0.01% BHT to 50 mg algal powder for extraction;

2. Vortex at room temperature for 20 min, centrifuge at 12,000 rpm, 4°C for 5 min, collect supernatant; re-extract pellet once and combine supernatants;

3. Concentrate, reconstitute in 150 µL dichloromethane, filter through 0.22 µm membrane, store in brown autosampler vials.

(2) Chromatographic conditions

1. Column: YMC C30 (3 µm, 100 mm × 2.0 mm i.d.);

2. Mobile phase: A = methanol/acetonitrile (1:3, v/v, with 0.01% BHT and 0.1% formic acid), B = methyl tert-butyl ether (MTBE, with 0.01% BHT);

3. Elution gradient as shown in Table 9;

4. Flow rate 0.8 mL/min, column temperature 28°C, injection volume 2 µL.

Time (min)	Mobile Phase B Ratio
0.00	0%
3.00	0%
5.00	70%
9.00	95%
10.00	0%
11.00	0%

Table 9. UPLC gradient elution program (carotenoid targeted)

(3) Mass spectrometry: atmospheric pressure chemical ionization (APCI), source temperature 350°C; curtain gas 25 psi; MRM mode acquisition; DP and CE individually optimized.

(4) Quantification: standard curves were generated with serial dilutions of standards (0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 5, 10, 25 µg/mL); external standard method was used for quantification. Sample content was calculated as:

Content (µg/g) = c × V / 1000 / m

where c is the concentration from the standard curve (µg/mL), V is the reconstitution volume (µL), and m is the sample weight (g).

4.6 Data Preprocessing and Statistical Analysis

1. Data preprocessing: missing values were imputed with 1/5 of the minimum value per row (metabolite); QC sample CV values were calculated and metabolites with CV < 0.5 were retained. Log2 transformation and centering were applied before PCA; UV (unit variance scaling) normalization was applied before clustering heatmap and K-means analysis;

2. Analysis included:

(1) Metabolite identification and quantification;

(2) Principal component analysis (PCA);

(3) Hierarchical cluster analysis (HCA) and clustering heatmap;

(4) Pearson correlation coefficient analysis between samples;

(5) Differential metabolite screening: fold change ≥ 2 or ≤ 0.5 (|log2FC| ≥ 1);

(6) Venn diagram of differential metabolites and K-means trend clustering;

(7) KEGG and MetMap functional annotation and enrichment analysis of differential metabolites (hypergeometric test, P-value) and differential abundance (DA) score analysis;

(8) Carotenoid content statistics, bar charts, and clustering heatmap analysis;

3. Analysis software: R 4.1.2 (ComplexHeatmap, corrplot, fmsb, MetaboAnalystR, impute, etc.).

qPCR Identification

This chapter covers the transcriptional validation workflow of *D. salina* transformants, including total RNA extraction, DNase digestion, cDNA synthesis, and SYBR Green RT-qPCR analysis. Detection targets include *CCD1*, *CitCCD4*, SP-*CCD1* fusion gene, and HygR (optional); *actin* was used as the housekeeping gene.

5.1 Experimental Design and Grouping

Experimental groups are shown in the table below. At least 3 independent biological replicates were set up per group, with 3 technical replicates per biological replicate. All samples were harvested on the same day under identical light conditions to avoid expression variation caused by light regulation of the *RBCS2* promoter.

Group	Construct	Promoter/Element	Detection Target
WT	Wild Type	—	*actin* (housekeeping)
EV	Empty Vector Control	—	*actin* / HygR
C1	p1302-*RBCS2*-*CCD1*	*RBCS2*	*CCD1*
C2	p1302-*RBCS2*-*CitCCD4*	*RBCS2*	*CitCCD4*

Table 10. qPCR experimental groups and detection targets

5.2 Total RNA Extraction (Vazyme Plant RNA Kit)

Total RNA was extracted using the Vazyme plant RNA miniprep kit (with on-column DNase). Strict RNase-free practice was followed throughout: RNase-free tips, tubes, and water; frequent glove changes; all operations on ice.

1. Take 5–10 mL log-phase algal culture, centrifuge at 4000×g, 4°C for 5 min to collect pellet; wash once with 0.5 M NaCl, centrifuge again and discard supernatant.
2. Immediately snap-freeze in liquid nitrogen, or add 450 µL lysis buffer (with β-mercaptoethanol).
3. Sonicate in ice bath (conditions same as sample prep), incubate at room temperature for 5 min.
4. Centrifuge at 12,000 rpm for 2 min, transfer supernatant to gDNA removal column, centrifuge to remove genomic DNA.

5. Add 0.5 volume of absolute ethanol, mix, transfer to RNA adsorption column.

6. Add DNase I working solution on-column, incubate at room temperature for 15 min.

7. Wash twice with wash buffer per manufacturer's instructions, spin empty column for 2 min to remove ethanol.

8. Elute with 30–50 µL RNase-free water, incubate at room temp for 2 min, then centrifuge to collect RNA.

9. Measure RNA concentration and purity with NanoDrop; A260/A280 ratio should be between 1.8 and 2.1.

5.3 cDNA Synthesis (Reverse Transcription)

Two-step reverse transcription was performed using the Vazyme PrimeScript FAST RT reagent kit:

1. gDNA removal system (16 µL): 1 µg total RNA + 4 µL 4× gDNA Eraser Premix + RNase-free water to 16 µL; incubate at 37°C for 10 min.

2. Reverse transcription reaction (20 µL): add 4 µL 5× RT Premix, mix gently.

3. Reaction program: 42°C for 15 min → 85°C for 5 s → hold at 4°C.

4. Dilute cDNA product 5–10× with RNase-free water; store at −20°C until use.

5.4 qPCR Reaction Setup and Program

SYBR Green quantitative PCR was performed using the Vazyme TB Green Premix Ex Taq II (Tli RNaseH Plus). Reaction system (20 µL):

Reagent	Volume
TB Green Premix Ex Taq II (2×)	10 µL
Forward Primer (10 µM)	0.8 µL
Reverse Primer (10 µM)	0.8 µL
cDNA Template	2 µL

Table 11. qPCR reaction system (20 µL)

Reaction program (three-step):

Step	Temperature	Time	Cycles
Initial Denaturation	95℃	30 s	1×
Denaturation	95℃	5 s	
Annealing/Extension	61℃	30 s	40×

Melting Curve	65℃→95℃	—	1×

Table 12. qPCR reaction program

5.5 Data Analysis (ΔΔCt Method)

1. Using *actin* as the housekeeping gene, calculate ΔCt = Ct(target gene) − Ct(*actin*) for each sample.
2. Using wild type (WT) as the calibrator, calculate ΔΔCt = ΔCt(sample) − ΔCt(WT).
3. Relative expression = 2^(−ΔΔCt).
4. Each experiment must include NTC (no template control), −RT control (no reverse transcriptase), and melting curve analysis to ensure primer specificity.

TLC Analysis

This chapter describes normal-phase silica gel thin-layer chromatography (TLC) for rapid screening of carotenoid cleavage products in transformant algae. Target products include β-ionone, β-apo-8'-carotenal, and β-citraurin and other aldehyde derivatives. TLC is a semi-quantitative screening method; positive results must be confirmed by HPLC-DAD or LC-MS.

6.1 Pigment Extraction and Sample Preparation

1. Take 1–5 mL algal culture, centrifuge at 4000×g for 5 min at 4°C to collect cells, wash once with 0.5 M NaCl.
2. Add 0.5 mL of acetone:methanol (1:1, v/v) mixture, sonicate in ice bath for 10 min for extraction.
3. Centrifuge at 12,000 rpm for 5 min, collect supernatant; re-extract pellet once and combine supernatants.
4. Add equal volume of n-hexane (pre-washed with saturated NaCl solution) to extract carotenoids; collect the upper organic phase.
5. Dry over anhydrous Na_2SO_4, evaporate under nitrogen, reconstitute in 50–100 µL acetone; store at −20°C protected from light.

6.2 TLC Chromatography Procedure

1. TLC plates: silica gel GF254 aluminum-backed plates (10×10 cm); activate at 105°C for 30 min before use, equilibrate in a desiccator.
2. Spotting: draw a light pencil line 1.2 cm from the bottom edge; spot with capillary tube, spot diameter ≤ 3 mm; standards 0.1–0.5 µg/spot, samples 5–15 µL.

3. Developing solvent: petroleum ether (60–90°C); pre-equilibrate the developing chamber for 20–30 min.

4. Development: remove plate when solvent front reaches ~1 cm from the top (approx. 8 cm); immediately mark solvent front with pencil, air-dry protected from light.

6.3 Visualization and Qualitative Interpretation

1. Observe carotenoid band color and position under visible light.

2. Observe GF254 fluorescence quenching bands under 254 nm UV light.

3. Iodine vapor staining: place plate in a sealed iodine chamber for 5–10 min to visualize all organic compound bands.

4. DNPH specific staining: spray with 0.4% DNPH (prepared in 2 M HCl), heat at 60–80°C for 5–10 min; aldehydes/ketones appear as yellow to orange-red spots. Aldehyde products such as β-citraurin and β-apo-carotenal are DNPH-positive, while β-carotene and lutein are negative.

5. Qualitative interpretation: a transformant is considered TLC-positive if it shows a new DNPH-positive spot absent in wild-type and empty vector controls, with Rf value matching the standard. Confirmation requires co-chromatography (1:1 mixed sample/standard spot shows no splitting) and at least 2 independent replicates.

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